SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation Supplementary Material

Anonymous CVPR submission

This supplement gives implementation details of SE(3)-WAM (Appendix A), pretraining on real robot data (Appendix B), the benchmark (Appendix C), the baselines, DOMINO comparison, real robot, and timing (Appendix D), additional results (Appendix E), the complete ablation study behind Sec. 4.4 (Appendix F), the error analysis that motivates the design (Appendix G), and additional related work (Appendix H). Colored dashes and bracketed entries denote values pending the final runs.

A. Implementation

A.1. Scene and Summary Tokens

Current image. A frozen MAE ViT-B/16 [19] encodes the 224×224 agent-view RGB into 196 patch tokens. We selected it over frozen DINOv3 encoders [42] and a CNN trained with the policy on development scenes (Appendix F.1). Each patch receives a view embedding, a linear embedding of its mean valid metric depth, and a linear embedding of the back-projected base-frame position of its center pixel, which we compute with the camera intrinsics and extrinsics. These positions also enter 3D rotary position embeddings for the patch queries and keys. The same center points are the source points of the sparse 3D flow (Appendix A.3). Target, robot, summary, and action tokens receive no rotary embedding. Past images never enter the model, and we drop their slots before the backbone, so the scene holds 212 tokens.

Target-motion history. At each query, the target history covers the last 8 control steps, the offsets $-350, -300, \ldots, 0$ ms. Each of the 8 target-state tokens is a 25-dimensional descriptor $(\mathbf{p}, \log R, \Delta\mathbf{p}, \log \Delta R, \Delta t, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \boldsymbol{\alpha})$ with dimensions $(3, 3, 3, 3, 1, 3, 3, 3)$. Slots 0–6 describe the seven intervals between consecutive offsets, and slot 7 holds the current absolute pose in the base frame. Velocities divide interval displacements by the elapsed time, and accelerations divide adjacent velocity differences by the interval-center spacing. We standardize each dimension with training statistics, zero the invalid dimensions, and append three validity flags (pose, interval, and acceleration) before a $28 \to 1024 \to 1024$ GELU MLP with an additive relative-time embedding. Invalid features are zeroed; a slot with no valid feature is masked out, without a learned lost-target token. In simulation, the poses come from the simulator. On the real robot, they come from the causal tracker of Appendix A.2, whose track we resample at the same offsets, interpolating linearly in translation and along the geodesic in rotation, without extrapolation and without interpolating across gaps longer than two camera periods.

Robot state. 8 end-effector states (position, rotation vector, and measured gripper opening fraction) at the same offsets form the robot tokens. The policy has no language token; view, time, depth, and position embeddings modify existing tokens rather than adding slots.

Summary tokens. 16 learned embeddings form the summary tokens of the world stream. They attend to the scene tokens and to each other at every layer, and they are the only world tokens that the action tokens attend to.

A.2. Pose Tracker

The real-robot protocol is fully marker-free, including online tracking, offline training labels, and error measurement. Each object has one fixed reference with a shared origin, axes, scale, and instance identity across demonstrations. The candidate pipeline uses causal masks from Grounding DINO and SAM 2.1 [30, 41] and RGB-D registration to this reference, with FoundationPose [48] as a candidate. [Validate and specify the tracker, reference acquisition, image resolution, and loss/recovery thresholds on the physical setup.]

Policy inputs must be obtained by causal replay of the deployment tracker. Each estimate carries its capture time, availability time, and component validity; later frames must not revise an estimate already issued to the policy. Offline supervision may use multiframe or multiview registration and manual review in the same object frame. These estimates are not independent ground truth. Tracker error measurements require a marker-free reference with stated uncertainty; reprojection consistency alone does not provide that reference.

For real-ball trials, the protocol excludes target orientation, relative rotation, angular velocity, and angular accelera-

Table S1. **Selected marker-free pipeline.** Median/p95 translation (mm) and rotation (°) error against a stated marker-free reference, its uncertainty, and the fraction of lost observations. Real-ball rotation is excluded. Measurements and the reference procedure remain pending; estimated labels are not ground truth.

Target	mm	°	Ref. uncertainty	Lost
Real rigid object	—	—	—	—
Real ball	—	–	—	—

tion from policy inputs, and target rotation from forecast supervision and reported errors. Ball-center translation, translational velocity and acceleration, and their valid observations are retained. The mask does not discard the full target-state slot or change robot orientation states, rotation commands, or the full SE(3) simulation experiments. No ball-orientation observability study is required. Table S1 records the selected pipeline's measurements when available. Actual error calibration and tracked closed-loop evaluation remain separate from the synthetic-noise study of Appendix F.2.

A.3. Forecast Heads

All forecast heads are light readouts of the final hidden states of the world stream. They are trained jointly with the action and skipped at deployment, and no token reads their outputs, so they shape the world stream only through the losses.

Target. A linear map from the mean final state of the 16 summary tokens reads out the trajectory $\{(\mathbf{d}_h, \boldsymbol{\phi}_h)\}_{h=1}^{8}$. With valid horizons $\mathcal{V}$,

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \left[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \right], \quad \text{(S1)}$$

with $\lambda_R = 0.1$ and $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^{\wedge}) R(\tau_t)$. We use the left-relative rotation vector and the cumulative translation instead of a twist exponential, which would couple translation to rotation through its Jacobian. Squared-error training yields a point estimate and does not represent multiple future modes explicitly.

Sparse 3D flow. The flow is the cumulative 3D displacement of the $N = 196$ patch center points over the 8 horizons, $F \in \mathbb{R}^{N \times 8 \times 3}$, in the base frame, together with a visibility $o \in \{0,1\}^{N \times 8}$. A linear readout on each patch token gives its point's displacement and visibility logit at every horizon. We divide the displacements by a fixed training-label scale $s_F = 0.7154605$ m and use

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h) \in \mathcal{V}_F} \left[\frac{\|\hat{f}_{ih} - F_{ih}/s_F\|^2}{3} + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \right], \quad \text{(S2)}$$

where $\hat{f}$ is the normalized displacement and $\mathcal{V}_F$ holds valid labels. Visibility BCE has weight 1 in simulation; invisible but valid points still supervise displacement. The points cover the target, the robot, and the context alike; the target's precision comes from its pose head.

Appearance targets. Linear readouts of the final patch tokens predict three targets at every horizon: metric depth and moving-surface RGB through 16×16 pixel readouts per patch, and 1024-dimensional features of a frozen DINOv3 ViT-L/16 teacher [42] on the 14×14 grid. Depth uses squared error on valid pixels; DINO features use squared error after channel-wise layer normalization; RGB uses squared error on surfaces that move by more than 1 mm, including the robot. These losses average channels and then valid positions. A $1024 \to 256 \to 3072$ MLP with GELU and a final sigmoid reads the mean of the final summary tokens to reconstruct the current target's 32×32 RGB region, using absolute error averaged over valid RGB elements. This ties the summary features to the target without adding an input. The four losses form $\mathcal{L}_{\mathrm{aux}}$; $\lambda_{\mathrm{aux}} = 0.1$ gives each a weight of 0.1 in the total objective. All four heads are used only in training and skipped at deployment; their outputs never enter the action stream.

A.4. Action Stream

Each chunk has 8 seven-dimensional commands, each with a base-frame translation increment, a left-relative rotation vector, and a gripper command, at a control period of 50 ms. Translation and nonzero rotation increments reference the measured end-effector pose at application; a zero rotation command can retain the controller's preceding orientation goal. Normalization uses the mean and the population standard deviation of valid, non-repeated issued commands in the training split. Flow matching uses a unit Gaussian in these coordinates with one u per chunk, and the action tokens receive u through adaptive layer normalization. Each prediction position draws 8 independent noise/time pairs for its chunk and averages their losses, which reduces the variance of the action gradient without enlarging the batch. At deployment, we integrate 5 Euler steps, reverse the normalization, decode the rotation vectors, and clip each axis to the exported motion limits: 0.05 m and 0.5 rad in the candidate configuration; 0.10 m translation requires an explicit deployment override. The gripper outputs 0 when $|g| \le b$ and otherwise $\mathrm{sign}(g)$, where $b = 0$ by default. The separately validated E5 deployment uses $b = 0.5$ for catching and $b = 0.1$ otherwise; it is not the final candidate's configuration.

A.5. Backbone and Coupling

Backbone. Each of the 24 layers applies one attention to all tokens with shared Q/K/V and output projections, fol-

lowed by separate SiLU feed-forward networks (ratio 4) for the world tokens and the action tokens. World tokens use layer normalization and action tokens adaptive layer normalization, for 504.0M trainable backbone parameters. The alternative of Appendix F.1 gives the two streams separate attention projections, normalization, and feed-forward networks at a matched parameter count.

Attention mask. The mask acts on three token groups, the scene ($\mathcal{S}$), the summary tokens ($\mathcal{M}$), and the actions ($\mathcal{A}$). Scene tokens attend only to $\mathcal{S}$; summary tokens attend to $\mathcal{S}$ and $\mathcal{M}$, and action tokens attend to $\mathcal{M}$ and $\mathcal{A}$. A blocked edge has the logit $-\infty$ at every layer and head, both in training and at deployment. Because the world tokens attend neither to the actions nor to the flow time, their states are computed once per query and reused across the Euler steps, and no future label reaches the action except through the gradients of the forecast losses on the shared parameters.

Screening couplings. The 4-layer variants of Appendix F.4 use the same modules, initialization, and losses and change only the mask or add a small module. R0 disables every forecast loss. R1 adds one object query and one query per patch to the world stream, which attend to $\mathcal{S}$, $\mathcal{M}$, and each other and carry the forecast readouts instead of the pooled states, but opens no edge between them and the actions. R2 lets the action tokens attend to these queries at every layer. R3 lets these queries attend to the actions, and R4 opens both edges; in R3 and R4, the world states are recomputed at every Euler step. R5 decodes the forecast once per query and encodes the target trajectory and the flow with visibility probabilities, with rotations in a continuous 6D form, through a geometry encoder (0.37M parameters) into geometry tokens that the action attends to in addition to $\mathcal{M}$. R5 trains on the predicted geometry and backpropagates the action loss through it without teacher forcing. We detach no gradient between the streams in any variant.

A.6. Training Recipe

Table S2 lists the recipe for training on Dynamic-14. Every control step of a demonstration is a query window, which gives 1,324,374 training windows from the 11,200 demonstrations, and 350 held-out episodes serve as the validation split. Each loss is normalized by its count of valid labels over the whole accumulated global batch. Each window contains 8 independent query positions with their own causal histories; there is no attention across these positions. The image encoder processes the current frame, and the DINOv3 teacher's features of the future frames are cached once per episode. The final simulation training is underway; its completed-run duration and hardware remain to be reported: [X] hours on [N] RTX PRO 6000 GPUs. The DROID base uses the recipe of Appendix B.4.

Table S2. Training hyperparameters on Dynamic-14.

Item	Value
Backbone	24 layers, width 1024, 16 heads, FFN 4096, 504.0M parameters
Tokens	212 scene, 16 summary, 8 action
Encoder	frozen MAE ViT-B/16, 224×224 input
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-4}
Learning rate	2×10^{-4}, 5,174 warm-up updates from 0, cosine decay to 0
Batch, updates	global batch 512 windows, 20 epochs (51,740 updates)
Loss weights	$\lambda_P = \lambda_F = 1$, $\lambda_R = 0.1$, $\lambda_{\text{aux}} = 0.1$ per target
Precision	trainable weights FP32, frozen encoder BF16, gradient clipping at 0.1
Action noise	8 noise/time pairs per query position
Sampling	5 Euler steps for the action
Inference	FP32, world states cached within a query
Hardware	[N] RTX PRO 6000 GPUs, about [X] hours per run

A.7. Deployment

Each query encodes the newest observation, runs the world stream once without its heads, and integrates the action stream on the cached world states. The chunk starts executing at the first control-period boundary at or after the query ends, and the next query starts at that same boundary with a new observation, so the policy replans as often as its latency allows. Under the real-time protocol of Appendix C.3, physics advances while a query runs and the executor continues with the remainder of the previous chunk. Appendix C.3 describes the executor shared by all methods, and Appendix D.4 the timing measurements.

B. Pretraining on Real Robot Data

B.1. Data, Partitions, and Time

The pretraining corpus is DROID [26], recorded at 15 Hz with two exterior cameras and one wrist camera, together with the 3D point tracks and optimized camera calibrations that PointWorld released for it [21]. We associate each released clip with its raw DROID episode by requiring exact agreement of the joint sequence, end-effector poses, gripper states, commands, and camera identity, and we exclude clips whose capture clock is not monotonic or whose association is ambiguous. Of 42,935 associated episodes, 42,087 yield usable windows. The model sees the first exterior camera, its depth, and PointWorld's calibration of it. The training unit is a causal observation window with its aligned future and action labels, 598,889 anchor windows in total: 538,740 from 37,874 training episodes and 60,149 from 4,213 validation episodes. The partitions share no episode. Table S3

Table S3. **DROID pretraining data.** Yield of each stage and validity of the labels. Windows are causal observation windows with aligned labels, one per control step of the main camera. Flow labels count point–horizon slots.

	Train	Val.	All
Episodes associated	–	–	42,935
Episodes with windows	37,874	4,213	42,087
Windows	538,740	60,149	598,889
Valid actions (%)	—	—	96.88
Valid flow labels (%)	—	—	47.05
Future frames for DINO	—	—	3,425,218
Windows with moving objects (%)	—	—	—

reports the yield and the validity of the labels.

Time. All times are capture timestamps, and we do not relabel frames to a nominal rate. The robot history uses the 8 most recent states at or before the query time, causally, as at deployment. Flow labels keep PointWorld's own record times rather than being interpolated to the 50 ms horizons of the simulation model. The future DINOv3 targets use the frames at the same record times. Each anchor expands into 8 consecutive native query positions with independent causal histories. Only the anchor position retains XYZ and DINO supervision; the remaining positions provide action supervision without interpolated or duplicated world labels.

Actions. We express robot states and Cartesian commands in the robot base frame after the calibrated flange-to-TCP transform, with the parameterization of Appendix A.4. We admit a label if its values are finite and its reference state was available causally. The robot's execution limits are not an admission rule, because a per-axis limit would discard fast but valid motion and bias the base toward slow demonstrations. Of 4,791,112 action slots, we mask the 149,318 at which DROID's collector issued no command and keep the remaining 96.88%, with no filter on magnitude or speed. Action and flow validity are independent, so an invalid action does not remove the flow labels of its window.

B.2. What the Base Learns

DROID provides no object poses, and its instructions name the manipulated object without marking it. The base therefore has no target-state tokens and no pose head. Its inputs are the current RGB-D frame and the robot history (204 scene tokens), with unavailable depth explicitly masked, and its objectives are the action loss (weight 1), the sparse 3D flow of the selected patch tracks (weight 1), and the future DINOv3 features (weight 0.1). Future depth, moving-region RGB, and the target reconstruction are off. Because PointWorld's tracks record only visible positions, the flow labels contain no negative visibility examples, and we set the visibility loss to zero rather than teach the model that every point stays visible. Most objects in DROID move only when the robot moves them, so the base mainly learns metric geometry, the 3D motion that the robot's own actions cause, and its coupling with the actions; the exogenous motion of our tasks and the target's pose enter during fine-tuning.

B.3. Flow Labels

PointWorld's tracks are already expressed in the base frame. Each patch of the 14×14 grid follows one fixed track over the window. Among the released tracks whose initial projection lies inside the patch, are initially visible, and have a valid positive depth, we prefer the one with the largest image motion and replace it by another candidate when its 3D displacement disagrees with its image motion. Otherwise we take the track nearest to the patch center. This rule covers moving objects that a fixed center point would miss: in a check of 30 segments from 10 episodes, 18 of which show visible object motion, it placed the start and two future points of a dynamic track inside the object in 13 of these 18 segments, against 5 for the center rule. This is coverage of image-consistent motion, not a measured 3D accuracy, and a patch on a small object can still drift to the background. The label is the track's displacement from its initial position at each record time. Patches without a valid track are invalid, and we do not fill them from neighbors. Over all windows, 47.05% of the point–horizon slots carry a valid label.

Label quality. We compared the PointWorld tracks with flow lifted from 2D correspondences [35] with monocular metric depth [46] on the same scenes. On surfaces that are static in the image, the selected PointWorld tracks move by [0.9] and [5.8] mm (median and 95th percentile) over the horizons, against [18] and [72] mm for flow lifted with monocular depth, whose scale drifts over time, so we use the PointWorld tracks and do not mix in the lifted flow. PointWorld's masks are not reliable enough to separate objects from the background, so we do not use them.

B.4. Recipe and Scale

Simulation model. SE(3)-WAM trains on Dynamic-14 from the generic visual initialization with the recipe of Table S2. We chose its depth among 12, 24, and 48 layers at width 1024 and its recipe with short runs at equal numbers of training samples, selected on development scenes (Fig. S2a).

DROID base. The base uses the same architecture family, with 48 layers at width 1024 (1.02B trainable parameters), and only real data. Its configured budget is 20 epochs (21,060 updates, 2,106 warm-up updates) at a global batch size of 512 on 64 GPUs, with the optimizer and precision of Table S2. Pretraining is ongoing; the final checkpoint is not yet available (Fig. S2b).

Figure placeholder: Fig. S1 — flow labels from PointWorld tracks (schematic)

Left: one DROID frame of the main exterior camera with PointWorld's optimized camera pose next to the raw one, and the 14×14 patch grid. **Middle:** in each patch, the candidate PointWorld tracks (thin gray) and the selected track (orange): the one with the largest image motion among the tracks that stay consistent in 3D, falling back to the track nearest the patch center; patches without a valid track greyed out. **Right:** the selected tracks' 3D displacements in the base frame at PointWorld's record times, with the robot's gripper path in purple. Use one real DROID frame sequence, vector overlays only.

Figure S1. Flow labels for the DROID base. Each image patch follows one PointWorld track, selected to cover moving objects where they exist, and its 3D displacement in the base frame is the label. Invalid labels mask their losses.

Figure placeholder: Fig. S2 — recipe and scale

(a) Simulation model. y: development SR; x: 12, 24, and 48 layers at equal sample exposure; the selected configuration starred. **(b) DROID base.** x: training samples (log); y: validation flow endpoint error at the last horizon (left axis, mm) and action loss (right axis) for the 48-layer run; marker at the selected checkpoint.

Figure S2. Recipe and scale of the simulation model (a) and the DROID base (b).

B.5. Task-Specific Fine-Tuning

Each real-robot policy is initialized from the base and fine-tuned on the demonstrations of its task only. The target-state tokens, the pose head, and the appearance heads that the base lacks are added with fresh initialization, and the policy receives poses from the tracker of Appendix A.2. Simulation weights do not initialize any real-robot policy, and the base does not see simulation data. Each policy creates its own optimizer, schedule, training-only normalization, and readouts.

C. Dynamic-14

C.1. Tasks, Roles, and Terms

We built Dynamic-14 on MuJoCo and robosuite [44, 60]. It contains fourteen tasks in five families (Fig. S3). Table S4 lists their success conditions and Table S6 their scenario ranges. We will release the tasks, the accepted scenarios, the success checks, and the demonstrations.

Roles and critical leads. Table S5 assigns the bodies of each task to the two roles of Sec. 3.1. The target is the body whose pose error enters a contact or containment tolerance ε, and the context consists of bodies that only need to be avoided with clearance m. A moving receptacle is therefore a target, since its pose decides containment. The wire of the passage tasks plays both roles. Its pose at the entry is a target, because the wand must be aligned within 30 mm there, and its geometry along the loop is context, because any contact fails. The table also gives each task's speeds and its critical lead $\Delta^\star = \varepsilon/\|\mathbf{v}\|$ (Appendix G.5), and its last column lists the error terms that each task exercises (Eq. 3 and Appendices G.2–G.3). Conveyor and translating targets set the speed and the lead. Rotating targets, SE(3) placement, and the wire make orientation enter the contact. Ramps, low friction, and strikers produce motion that extrapolation misses, the occluder, wire, tray, and striker set the clearance, and the occluder also removes history.

We check success at every 2 ms physics step, after the ordered requirements of Table S4, and an interrupted continuous condition resets its counter. Terminal events fall into five categories that we report separately. An *invalid* event is a corrupted simulator state. *Boundary* events are a workspace or downstream exit, a drop, or an expired goal window. *Safety* events are unsafe contact, an excessive rim impulse, wire contact, a release impulse, early robot contact, unsupported tray contact, or a striker trap. *Task violations* are a release outside the goal, a path deviation or backtracking, a premature release, a deflection without capture, tipping, a missing or extra impact, or an insufficient change of outgoing motion. The last category is the *timeout*. Each family evaluates its events in a fixed order per physics step and keeps the first terminal event. The task-balanced score weights every task equally, including tasks on which some methods never succeed.

C.2. Scenarios and Demonstrations

We sample continuous ranges uniformly and discrete choices with equal probability, and then apply physical-feasibility checks. Evaluation replays the stored accepted scenarios. Harmonic translation follows $\mathbf{p}(t) = \mathbf{c} + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0)$ with $\phi \sim \mathcal{U}[-\pi, \pi]$. Capture targets start at rest and move under gravity and friction on ramps of $[5, 15]^\circ$. We accept an occluded scene only if the target is visible for at least eight initial frames, fully occluded for 0.5–3.0 s, and visible again for at least three frames.

The training pool contains 11,200 demonstrations, 800

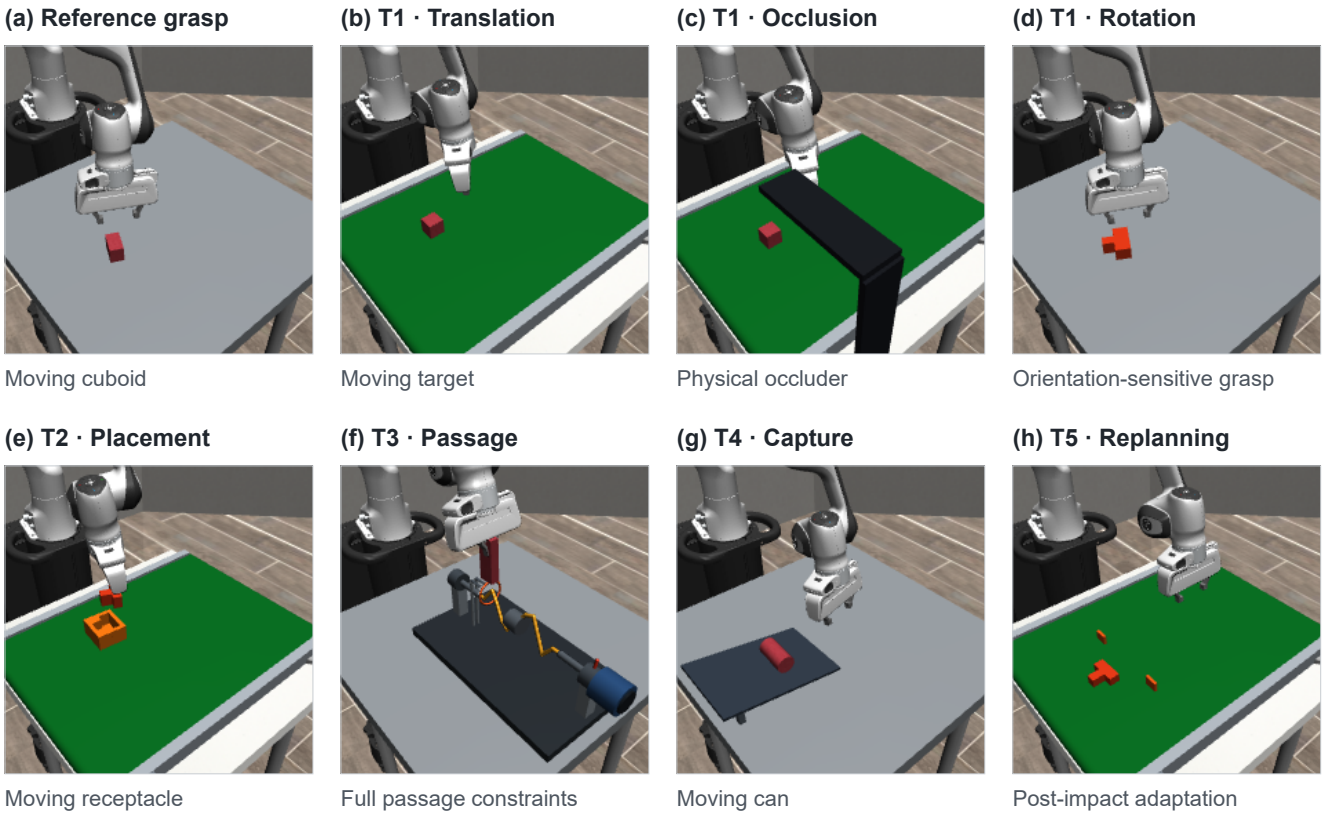


Figure S3. Dynamic-14 observations. Eight examples cover reference grasping, grasping a translating, occluded, or rotating target, placement into a moving receptacle, passage along a rotating wire, capture of a rolling object, and grasping after an impact. Each panel is the final observed, uncropped 224×224 agent-view RGB of one development scene. Table S4 lists all fourteen tasks.

per task. They were generated by experts with privileged simulator state, whereas the learned policies receive only the inputs of Sec. 3.2. We record the demonstrations at 20 Hz with camera images, depth, robot states, issued commands, and simulator ground truth for target poses and scene geometry. From these we compute the future labels, their validity masks, and the command labels. The policies receive the target pose from the simulator, which the pose-noise sweep perturbs (Appendix C.3).

C.3. Protocol and Sweeps

Executor. Every method runs under the same asynchronous executor on a 50 ms control clock, and queries start on its period boundaries. The next query starts at the first boundary at or after the end of the previous inference, which is also when the chunk of that inference starts executing. While a query runs, the executor continues with the remainder of the previous chunk, one command per control period, and the new chunk replaces this remainder. A method whose inference fits in one control period therefore replans before every command, and a method with latency ℓ replans every $\max(1, \lceil \ell / 50\,\mathrm{ms} \rceil)$ control periods. Before the first chunk of an episode arrives, and whenever a chunk runs out before the next one, the controller holds the current arm target and gripper command while physics advances; no command is repeated. A baseline whose paper replans less often keeps its own chunk length under the *original executors* protocol.

Real-time protocol. Unless noted otherwise, physics advances during inference by a latency drawn per query from the method's measured latency distribution on one RTX 5090 (Appendix D.4). The post-impact grasp additionally has a 50 ms command delay built into the task. Table S14 reports three alternatives. With *paused* physics, the shared executor is kept and the simulation waits for inference; any added artificial delay still advances physics. With *equal latency*, every method is charged the same latency, the p95 of SE(3)-WAM, regardless of its own speed. With *original executors*, each baseline keeps the execution scheme of its paper.

Latency. An added delay of 0, 50, 100, 150, or 200 ms is inserted after inference on top of the protocol's latency, and physics advances during it.

Table S4. Success conditions of the fourteen Dynamic-14 tasks, with the thresholds of our task implementations. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

Task	Ordered success conditions	Limit
grasp; t1_belt, t1_xyz, t1_so3, t1_occ	Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC.	160 steps 8 s
t2_trans	Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both ≥ -0.002 m, top no higher than the opening. Release requires an opening command < -0.5 and no fingerpad contact. Relative linear/angular speeds ≤ 0.08 m/s and 0.8 rad/s for 0.10 s, followed by a 0.50 s stable dwell.	200 10 s
t2_se3	Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance ≥ 0.002 m, floor gap in $[-0.002, 0.002]$ m, tilt ≤ 0.05 rad, top below opening. After release, relative speeds ≤ 0.02 m/s and 0.04 rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach.	200 10 s
t3_phase	The episode starts with the wand already grasped. Phase alignment: distance ≤ 0.090 m and loop-direction error ≤ 0.20 rad. Enter within 0.030 m at progress ≤ 0.15; reach historical maximum progress ≥ 0.60, then ≥ 0.85 and endpoint distance ≤ 0.055 m.	420 21 s
t3_full	Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress ≥ 0.85 and move at least 0.045 m from both path and endpoint. Finally open (< -0.5) with no fingerpad contact. Releasing before exit fails.	640 32 s
t4_sphere	Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s.	120 6 s
t4_sphere_tabletop; t4_slider	Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed ≥ 0.08 m/s. Bilateral hold for 0.05 s, COM lift ≥ 0.05 m from the first attainment of that hold criterion, then contact and height for 0.50 s.	120 6 s
t4_can	Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical > 0.45 fails as unstable tipping. The configuration does not require a transverse closing axis.	120 6 s
t5_replan	Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed ≥ 0.035 m/s and yaw rate ≥ 0.35 rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift ≥ 0.05 m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s.	120 6 s

Pose noise. With frozen weights, we add three synthetic conditions: nominal temporally correlated pose noise, twice its amplitude, and nominal noise with contiguous missing observations. Perturbations affect current and historical estimates before caching and kinematics, with the clean real-time result reused as reference (Appendix F.2). These conditions are not calibrated to a physical tracker; actual calibration and tracked closed-loop evaluation remain pending.

C.4. Splits and Statistics

We select final configurations on development scenes. The main Dynamic-14 comparisons and final-model ablations use a separate held-out set of 1,400 scenes, generated with the same ranges and with new source seeds that appear in no training, validation, or development episode. The first 100 feasible scenes in increasing seed order form each task's set. The screening results in Appendix F retain their original splits and execution protocols.

The full model, CV-input variant, and core ablations in Table 3 are assigned training seeds 0, 1, and 2. Screening studies retain their recorded seed counts. Baselines use a single training seed (Table S7). On development scenes, three training seeds of the same configuration differed by up to [7.4] points in SR. Most of this spread came from the wire passage, whose success ranged from [0] to [71]% across seeds, and we do not interpret single-seed differences of a few points. We report each training seed and the cross-seed mean and standard deviation. We also average the seeds per scene and bootstrap held-out scenes within each task, jointly across methods, for paired 95% intervals. These intervals quantify scene uncertainty conditional on the trained weights; training-seed variation is reported separately. We

Table S5. **Roles, tolerances, and critical leads of the Dynamic-14 tasks.** The target is the body whose pose error enters a contact or containment tolerance ε. The context comprises bodies that must only be avoided, with clearance m. Speeds follow the sampling ranges of Table S6 (capture tasks: at the ramp exit), and the critical lead $\Delta^\star = \varepsilon/\|\mathbf{v}\|$ is given for the median speed. Terms: error terms that a task exercises (Eq. 3 and Appendices G.2–G.3). v: moving target. $r\theta$: orientation enters the contact. e_{dyn}: non-constant motion (harmonic, gravity, friction, impact). m: a context body must be avoided. Occ.: history gaps. Delay: command delay built into the task.

Task	Code	Target	Context	ε (mm)	$\|\mathbf{v}\|$ (m/s)	$\Delta^\star$ (ms)	m (mm)	Terms
Harmonic grasp	grasp	block	table	[X]	0.03–0.20	[X]	–	v, $r\theta$, e_{dyn}
Conveyor grasp	t1_belt	part	belt	[X]	0.04–0.10	[X]	–	v
Translating grasp	t1_xyz	part	table	[X]	0.04–0.15	[X]	–	v, e_{dyn}
Rotating grasp	t1_so3	part ($\omega \leq 1.2$ rad/s)	table	[X]	0.04–0.15	[X]	–	v, $r\theta$, e_{dyn}
Occluded grasp	t1_occ	part	occluder canopy	[X]	0.04–0.10	[X]	[X]	v, m, occ.
Moving receptacle	t2_trans	receptacle	table	[X]	0.03–0.06	[X]	–	v
Rotating receptacle	t2_se3	T-shaped receptacle	table	[X]	0.03–0.06	[X]	–	v, $r\theta$
Wire passage	t3_phase	wire loop (entry)	rotating wire	30	[X]	[X]	swept	v, $r\theta$, m
Wire passage, full	t3_full	wire loop (entry)	rotating wire	30	[X]	[X]	swept	v, $r\theta$, m
Ball capture	t4_sphere	ball	tray, ramp	[X]	[0.4–0.7]	[X]	[X]	v, e_{dyn}, m
Ball capture, table	t4_sphere_tabletop	ball	table, ramp	[X]	[0.4–0.7]	[X]	–	v, e_{dyn}
Slider capture	t4_slider	slider	table, ramp	[X]	[0.4–0.8]	[X]	–	v, e_{dyn}
Can capture	t4_can	can	table, ramp	[X]	[0.4–0.7]	[X]	–	v, $r\theta$, e_{dyn}
Post-impact grasp	t5_replan	part	striker	[X]	0.12–0.14	[X]	[X]	v, $r\theta$, e_{dyn}, m, delay

treat differences whose interval lies within ± 2 points as ties.

C.5. Metrics

The task-balanced success rate is

$$\mathrm{SR}(m) = \frac{1}{14}\sum_{k=1}^{14}\frac{1}{n_k}\sum_{i=1}^{n_k} y_{mki}, \tag{S3}$$

for binary outcomes y_{mki} of method m on scene i of task k. The *feasible contact window* of a scene is the set of control-period boundaries t at which the privileged expert could make the contact of the task's first success stage. A time t belongs to it when the expert's contact frame on the target at t, taken from a rollout of the same scene with the robot held at its initial pose, (i) satisfies the geometric conditions of that stage (Table S4), (ii) has an inverse-kinematics solution within the expert's joint limits, and (iii) can be reached from the initial end-effector pose by t under the expert's per-command limits of 0.05 m per translation axis and 0.5 rad per rotation-vector component. The earliest such time is the larger of $\lceil \max_i |\Delta p_i|/0.05\,\mathrm{m} \rceil$ and $\lceil \max_i |\Delta\phi_i|/0.5\,\mathrm{rad} \rceil$ control periods, a lower bound that ignores the controller's tracking lag. Only robot–target collisions are disabled in this offline fixed-robot reference replay; target–environment collisions remain enabled, and policy rollouts retain their original physics. The window depends only on the scene and is shared by all methods. The two placement tasks start with the object already held, so their contact is the first contact of the held object with the goal receptacle, and their contact frame is the end-effector pose implied by the expert's placement target. The phase-only threading task (`t3_phase`) also starts with the wand held and succeeds without any contact, so it has no contact errors; their averages cover the other 13 tasks, and SR covers all 14. The *contact-time error* is the distance from the first-contact time to this window, and it is zero inside the window. The *position error at first contact* is the distance of the end effector from the contact frame at that time. We report both on episodes that make contact. Failures without contact stay in the SR denominator, and their contact error is missing rather than zero. *Collisions* count safety events. *Time-to-action* is the wall-clock time from the arrival of an observation to the availability of the first command (Appendix D.4), and the *action age* is the execution time minus the capture time.

Table S6. Scenario proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

Task	Initial-state proposal	Motion or event parameters
grasp	Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$.	$v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$.
t1_belt	Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist.	Belt $v_y \in [.04, .10]$.
t1_occ	Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$.	Same belt range; clearance parameter .20; physical occlusion.
t1_xyz	Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$.	$v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$.
t1_so3	Same as XYZ.	Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities.
t2_trans	Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation.	Goal $v_y \in [.03, .06]$; no rotation.
t2_se3	Same goal x, y range, $z = .804$; yaw $[-.3, .3]$.	Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities.
t3_phase	Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$.	Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values).
t3_full	Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range.	Same wire rotation; cradle and release requirement enabled.
t4_sphere	Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$.	Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture.
t4_sphere_tabletop	Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$.	Same slope and release; target/ramp friction $[.7, .9]$, table friction .35.
t4_slider	Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp.	Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$.
t4_can	Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent.	Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$.
t5_replan	$x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist.	Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms.

D. Baselines, External Benchmarks, Real Robot, and Timing

D.1. Baselines

Common conditions. Table S7 summarizes the adaptations. Each baseline keeps its official architecture, inputs, action chunk, and initialization. ACT and Diffusion Policy train from scratch, $\pi_{0.5}$ is fine-tuned from `pi05_base`, and the other methods start from their released weights. All of them train on the same 11,200 demonstrations, whose commands we map to each action space through the same Cartesian controller. Dynamic-14 renders a 224×224 wrist camera in addition to the agent view, and methods whose recipes use a wrist view receive it. SE(3)-WAM and its CV-input variant use the agent view only. All methods run under the shared executor and the real-time protocol of Appendix C.3, and Table S14 adds their original executors.

Training. Each baseline follows its official recipe for the published number of updates and is trained once, with seed 0. We changed only the data loading and the action space. For DreamVLA, we ran both its original 40-epoch schedule and a 20-epoch cosine schedule and report the one that was better on development scenes. The configuration of SE(3)-WAM, in contrast, was selected by the ablations of Appendix F, a search that the baselines do not receive, and the full model, CV-input variant, and final-model core controls are each assigned three training seeds. Screening studies retain their group-specific seed counts.

Inputs. PUMA [14] receives its historical optical flow, computed on the rendered frames. DreamVLA receives seven causal RGB frames, the end-effector position and orientation, the finger positions, and a fixed task text, and it samples with ten DDIM steps and classifier-free guidance of 1.5. OpenWAM-α receives the current RGB views, joint values, gripper opening, and task text, and it generates video and actions jointly with video decoding disabled. DynamicWAM receives its flow history and kinematic descriptors.

SE(3)-augmented baselines. $\pi_{0.5}$+SE(3) and DynamicWAM+SE(3) receive the eight target-state tokens of Appendix A.1 from the same pose source, with the same validity flags and the same masking of slots without valid features. A two-layer MLP maps the tokens to the model's width. For $\pi_{0.5}$, we append them to the prefix after the image tokens, where the action expert attends to them, and for DynamicWAM they join its kinematic descriptors. The output layer of the MLP is zero-initialized, and training follows the base model's recipe. This initializes the added embeddings to zero; it does not guarantee unchanged attention or actions. These controls match target-state access, while the depth-input difference remains.

CV-input control. *SE(3)-WAM, CV input* shares the scene tokens and the action stream of SE(3)-WAM but has no forecast losses. Instead, the action stream receives constant-velocity target-pose extrapolations at the chunk's estimated execution times as input tokens. These estimates use only history and a delay estimate available at query time, never a future pose or the realized latency of that query. Training follows the same data and recipe and retains current-target RGB reconstruction. The no-future control in Table 3 has no extrapolated input tokens. Together these comparisons distinguish added trajectory inputs from future supervision; the CV-input model is a separately trained policy.

D.2. DOMINO

The external comparison uses DOMINO [14] and four methods: SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA. They share the selected single-arm Franka tasks, demonstrations, and evaluation scenes, and use the benchmark's native evaluator. The planned comparison uses the 24 prepared clean-L1 tasks, one training seed (0) per method, and 50 episodes per task and method, for 4,800 episodes. We report SR and manipulation score (MS). The task list, action semantics, and control timing are checked before formal training; test scenes are separate from interface debugging and model selection. [Complete the validated DOMINO protocol and results.] SE(3)-WAM uses domain-specific normalization and the simulator pose of the target selected by the instruction; missing target-state slots are masked as in Appendix A.1. This privileged input is disclosed separately from the RGB-based baselines. The DOMINO comparison adds no component ablations, pose-noise conditions, or additional latency sweep.

D.3. Real Robot

Platform. The robot is a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera, commanded through [a timestamped Cartesian interface] at [control frequency]. The camera records RGB-D at [resolution] and [frame rate], and we calibrate its extrinsics by [procedure] before each session. SE(3)-WAM uses this fixed third-person view; baselines whose recipes require a wrist view receive [per-method camera configuration]. All policies run on one RTX 5090.

Tasks and success. Table S9 defines the six tasks. T1 and T2 share a belt conveyor, T3 uses a turntable, T4 a remotely driven toy car, T5 a cue-driven collision between two balls, and T6 a ramp. Each task has its own operating range and success condition. Touching or pushing the target without completing the task, a single-finger hold, and an action after the target has left the workspace count as failures; [equipment-invalid trials and safety interventions are repeated or counted as failures by the following rule].

Trials. We compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and Diffusion Policy [12]. Each method runs 20 trials per task. All methods use the same pre-generated trial keys, which fix the object instance, the initial pose, and the motion parameters, sampled by [sampling protocol]. Trials run in [randomized method order], so that drift of the apparatus does not align with the methods. An operator resets the scene. A trial starts at the first usable observation, and the robot may move at any time; a trial ends at success, failure, or a timeout of [X] s. Demonstrations, development trials, and test trials come from separate sessions; test initial conditions are sampled independently under the same task protocol.

Training and adaptation. SE(3)-WAM is fine-tuned per task from the DROID base (Appendix B.5). The baselines are trained or fine-tuned on the same demonstrations, with [per-method initialization and training settings]. The marker-free tracker and offline labels share one fixed object reference (Appendix A.2); [specify the selected reference acquisition and registration settings].

Analysis. Table S10 reports successes per task, 95% binomial intervals for the pooled rates, and paired differences from a bootstrap over trial keys. The evaluation took [X] days.

D.4. Timing

We time all methods on one RTX 5090 (32 GB), the target deployment GPU of the robot. For each method, each of three

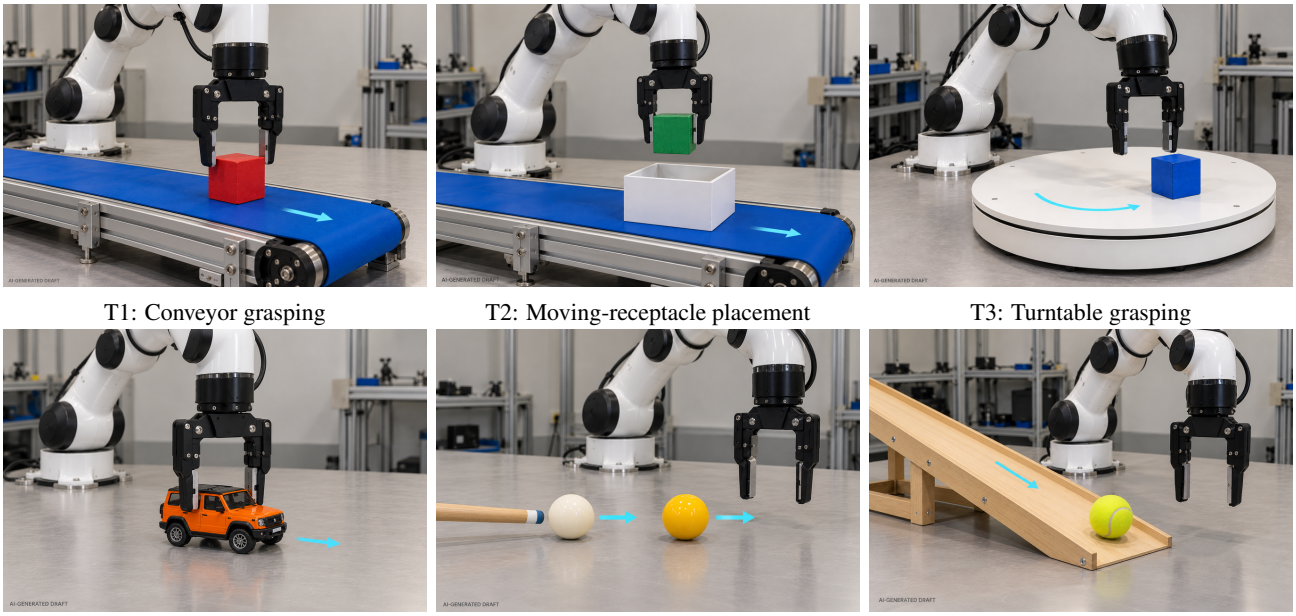
Table S7. **Baseline adaptations.** All methods train on the same 11,200 demonstrations with the inputs of their papers and run under the shared executor of Appendix C.3. Updates × batch follows the official recipe. Exec.: the execution scheme of the original paper (commands executed per chunk before replanning), used only for the original-executor control of Table S14. Params: trainable parameters, frozen encoders excluded. Seeds: planned training seeds; completion is verified before filling results.

Method	Initialization	Inputs	Updates × batch	Exec.	Params	Seeds
ACT [57]	scratch	agent + wrist RGB, robot state	$50{,}000 \times 32$	temporal agg.	[X]	1
Diffusion Policy [12]	scratch	agent + wrist RGB (published horizon), robot state	$50{,}000 \times 32$	8	[X]	1
$\pi_{0.5}$ [38]	`pi05_base`	current agent + wrist RGB, robot state, task text	$30{,}000 \times 256$	10	[X]	1
$\pi_{0.5}$ + SE(3)	`pi05_base`	as $\pi_{0.5}$ + SE(3) motion tokens	$30{,}000 \times 256$	10	[X]	1
DynamicVLA [50]	released DOM ckpt.	frames $[-2, 0]$, robot state, task text	$50{,}000 \times 44$	streaming	[X]	1
PUMA [14]	official checkpoint	RGB, optical-flow history, robot state, task text	[X]×[X]	[X]	[X]	1
DreamVLA [55]	released components	seven causal agent + wrist RGB frames, robot state, task text	$48{,}220 \times 512$	1	[X]	1
OpenWAM-α [47]	official foundation	current agent + wrist RGB, joints, gripper, task text	$34{,}490 \times 192$	32	[X]	1
DynamicWAM [33]	official initialization	RGB, flow history, kinematic descriptors	$40{,}000 \times 128$	16	[X]	1
DynamicWAM + SE(3)	official initialization	as DynamicWAM + SE(3) motion tokens	$40{,}000 \times 128$	16	[X]	1
SE(3)-WAM, CV input	as SE(3)-WAM	as SE(3)-WAM, plus constant-velocity forecast tokens, no forecast losses	as SE(3)-WAM	–	[X]	3
SE(3)-WAM	frozen MAE encoder	current agent RGB-D, SE(3) and robot histories	$51{,}740 \times 512$	1	523.8M	3

Table S8. **DOMINO external evaluation.** Planned comparison on 24 single-arm clean-L1 tasks: one training seed per method and 50 episodes per task and method (4,800 episodes total). SR (%) is grouped by interaction type and averaged over tasks; MS is the manipulation score. Protocol: Appendix D.2.

Method	Intercept	Track	All	MS
$\pi_{0.5}$	—	—	—	—
DynamicWAM	—	—	—	—
PUMA	—	—	—	—
SE(3)-WAM	—	—	—	—

rounds uses thirty warm-up calls and 300 measured calls over three observation sequences of eight frames, with CUDA synchronization at call boundaries and rotated method order. Each method uses the precision of its release; SE(3)-WAM uses FP32. The time-to-action includes encoding the image and generating the first command, and on the robot also tracking the newest frame. For SE(3)-WAM, we record the time of the forecast heads separately, although deployment skips them. Table 1 gives the p95 values and Table S7 the parameter counts. On the robot, traces record capture, availability, preprocessing, the start and end of inference, queue entry, command issue, and execution. The action age is the execution time minus the capture time, and we report its p50, p95, and maximum. A query misses its deadline when its time-to-action strictly exceeds the 50 ms control period; the next period then uses a remaining command or holds the current target if none is available. The fraction includes all queries, including the first query of every trial; hold-only periods do not contribute an action age. We also report peak memory (Table S11). The latency sweep of Dynamic-14 differs from wall-clock deployment, because it inserts a prescribed delay while physics advances.



T1: Conveyor grasping | T2: Moving-receptacle placement | T3: Turntable grasping

T4: Moving-car grasping | T5: Post-collision interception | T6: Ramp interception

Figure S4. **Real-robot tasks.** AI-generated concept illustrations, to be replaced by photographs of the setup; arrows indicate the intended target motion. T6 shows the rolling variant; the task also includes sliding objects.

Table S9. Real-robot task definitions. $N = 100$: training demonstrations per task; 20 additional demonstrations per task are reserved for validation.

Task	Device, motion, and objects	Operating range; N	Success and failure criteria
T1	Conveyor transports a graspable cube.	[belt speed]; 100	[stable grasp and lift; failures]
T2	Conveyor transports an open box; the arm places a cube into it.	[belt speed]; 100	[placement and retention; failures]
T3	Turntable carries a cube around its axis.	[angular speed, radius]; 100	[stable grasp and lift; failures]
T4	Remotely driven toy car moves on the tabletop.	[speed, path]; 100	[car grasp and lift; failures]
T5	A cue-driven ball strikes a second ball; the arm intercepts the second ball after the collision.	[impact and outgoing speed]; 100	[post-impact capture; failures]
T6	A ball rolls or a block slides down a ramp toward the interception region.	[slope, release, speed]; 100	[capture and retention; failures]

Table S10. **Real-robot successes** per task (successes/trials, 20 trials per task; tasks in Table S9). All: pooled success rate (%) with its 95% binomial interval. Δ: paired difference from SE(3)-WAM (points) with a bootstrap interval over trial keys.

Method	T1	T2	T3	T4	T5	T6	All	Δ
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—	–

Table S11. **Latency breakdown** of one query of SE(3)-WAM on one RTX 5090 at batch size 1 (ms, p50 / p95), and action age on the robot.

Stage	p50	p95
Pose tracking (per frame, robot)	—	—
Image encoding	—	—
World stream (once per query)	—	—
Action stream (5 Euler steps)	—	—
Time-to-action (total)	—	—
Forecast heads (training only, for reference)	—	—
Action age at execution (robot)	—	—
Queries over 50 ms (robot, %)	—	

E. Additional Results

E.1. Per-Task Results and Outcomes

Table S12 breaks the held-out Dynamic-14 results of Table 1 down by task for every method, with the paired interval of SE(3)-WAM against the strongest other method on each task. Table S13 reports how episodes end. A method that trades one failure for another, e.g., fewer collisions for more timeouts, shows this only here. Every failure stays in the success-rate denominator.

E.2. Protocols and Latency

Table S14 evaluates every method under three further protocols. Paused physics removes the cost of computation. Equal latency charges every method the p95 latency of SE(3)-WAM and so separates a method's speed from the way it uses time. The original executors test whether our shared executor puts any method at a disadvantage. With paused physics, the difference between SE(3)-WAM and [DreamVLA] is [$\pm$X.X] points, and under equal latency it is [$\pm$X.X] points. The original executors change the SR of the baselines by at most [X.X] points. Table S15 gives success at every added latency of the sweep, and its rows for SE(3)-WAM and the strongest baseline are the data of Fig. 1 (right).

E.3. Negative and Null Results

Screening results. The history, forecast, encoder, and coupling comparisons use the screening recipes of Appendix F. Their original splits and seed counts must accompany each result. Q3 coupling effects changed sign across seeds; this does not establish final-scale equivalence between training-only supervision and online forecast feedback. The new four-control study reports signed differences and paired intervals in Table 3. [Summarize completed null and negative effects at their actual scale.]

The completed history and forecast screening results are reported at their original scale in Tables S17 and S20. Table S18 reports the stationary-history intervention on the frozen screening checkpoint; it is not a final-scale training ablation.

Unsolved tasks. The SE(3) placement into the rotating receptacle (t2_se3) fails for every method (Table S12). [The policies reach the receptacle but miss its angular tolerance before it turns away.] No variant of SE(3)-WAM solves the full wire passage (t3_full), which adds grasping and releasing the wand to the passage, whereas [OpenWAM-α] succeeds in [XX]% of these episodes. On the passage that starts with the wand grasped (t3_phase), the success of SE(3)-WAM ranges from [0] to [71]% across training seeds.

Other failures. The collapse of Fig. S6a is weakest for the capture tasks (Spearman $\rho = [X.XX]$), where success also depends on when the gripper closes. The three synthetic-noise conditions change SR by [report the three signed differences from the clean reference]; these results are separate from physical-tracker calibration. Most remaining failures of SE(3)-WAM are [e.g., timeouts after a missed first grasp] (Table S13).

E.4. Forecasts at Execution Time

Fig. S5 aligns the query, the forecast, the ground truth, the issued actions, and the first contact on physical timestamps. The examples show where forecast errors arise—e.g., at the moment of an impact or when a rolling object leaves its ramp—alongside contact-time and pose errors. These aligned diagnostics do not establish that the deployed action stream reads the decoded forecasts or that a forecast error caused a particular action error.

E.5. Supplementary Video

The supplementary video shows real-robot trials of SE(3)-WAM and the strongest baseline on all six tasks with the tracked poses overlaid, a set of Dynamic-14 episodes per task family with the forecasts of the training-time heads, and the failure cases discussed in Sec. 4.3.

Table S12. **Per-task held-out success** (%) on Dynamic-14 for every method of Table 1, mean of three training seeds for our variants and one for baselines, real-time protocol. DP: Diffusion Policy. DVLA: DynamicVLA. DrVLA: DreamVLA. OWAM: OpenWAM-α. DWAM: DynamicWAM. +: with our SE(3) history tokens. CV: SE(3)-WAM with CV input. σ: SE(3)-WAM with nominal synthetic temporally correlated pose noise under the approved robustness setting, reusing those episodes. Δ: SE(3)-WAM minus the strongest other method on that task, with its 95% paired interval. Task codes are listed in Table S5.

Task	ACT	DP	$\pi_{0.5}$	$\pi_{0.5}$+	DVLA	PUMA	DrVLA	OWAM	DWAM	DWAM+	CV	σ	**Ours**	Δ [95% CI]
Harmonic grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Conveyor grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Translating grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Rotating grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Occluded grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Moving receptacle	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Rotating receptacle	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Wire passage	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Wire passage, full	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ball capture	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ball capture, table	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Slider capture	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Can capture	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Post-impact grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Task-balanced	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Figure placeholder: Fig. S5 — forecasts aligned with what happened

Rows 1–4: the median-error held-out episode of each motion type (conveyor, rotating target, rolling capture, post-impact); **row 5:** a failure. **Columns:** the query frame at τ_t; the frame at the first command's execution time τ^{a}; and a 3D inset in the base frame. **Overlays:** blue past SE(3) poses; orange predicted target triads at $\tau_t + \delta_h$ and sparse flow arrows, read from the training-time heads; gray ground-truth poses at the same physical times; purple issued gripper path; a marker at first contact. A strip under each row plots translation and rotation error vs. horizon for learned SE(3), constant velocity, and constant acceleration. Annotate the failure cause of row 5 in red.

Figure S5. Forecasts of the training-time heads on held-out episodes, aligned in physical time with the ground truth and the issued actions. Rows 1–4 show the median-error episode of each motion type, and row 5 shows a failure.

Table S13. **Outcome categories** on held-out Dynamic-14 scenes (% of episodes, seed-averaged, real-time protocol). Categories follow Appendix C.1, and rows sum to 100. The noisy-pose row reuses the nominal synthetic correlated-noise episodes from the robustness study.

Method	Success	Safety	Task	Bound.	Timeout	Inval.
ACT	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—
$\pi_{0.5}$	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—	—
DynamicVLA	—	—	—	—	—	—
PUMA	—	—	—	—	—	—
DreamVLA	—	—	—	—	—	—
OpenWAM-α	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—	—
SE(3)-WAM, CV input	—	—	—	—	—	—
SE(3)-WAM, noisy poses	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—

Table S14. **Protocol controls** on held-out Dynamic-14 scenes (SR, %). Paused: shared executor, with physics stopped during inference. Equal latency: shared executor, with every method charged the p95 latency of SE(3)-WAM. Original executor: real time with each method's own execution scheme. Real time: the protocol of Table 1. Late: queries in real time whose time-to-action exceeds the 50 ms control period (%).

Method	Paused	Equal latency	Original executor	Real time	Late (%)
ACT	—	—	—	—	—
Diffusion Policy	—	—	—	—	—
$\pi_{0.5}$	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—
DynamicVLA	—	—	—	—	—
PUMA	—	—	—	—	—
DreamVLA	—	—	—	—	—
OpenWAM-α	—	—	—	—	—
DynamicWAM	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—
SE(3)-WAM, CV input	—	—	–	—	—
SE(3)-WAM	—	—	–	—	—

Table S15. **Success versus added latency** on held-out Dynamic-14 scenes (SR in %, real-time protocol). The 0 and 150 ms columns equal the All and +150 ms columns of Table 1. Slope: least-squares change of SR per 100 ms of added latency (points).

	Added latency (ms)					
Method	0	50	100	150	200	Slope
ACT	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—
$\pi_{0.5}$	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—	—
DynamicVLA	—	—	—	—	—	—
PUMA	—	—	—	—	—	—
DreamVLA	—	—	—	—	—	—
OpenWAM-α	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—	—
SE(3)-WAM, CV input	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—

F. Core Ablations and Screening Studies

Final-scale confirmation. Table 3 contains four independently trained controls of the 24-layer final model, with three training seeds each. They keep the final data, initialization recipe, training exposure, and other components fixed (Appendix A.6). Evaluation uses 1,400 held-out scenes under the real-time protocol at 0 and 150 ms added delay, with shared scene keys and inference seeds. The full-model reference is reused from the main evaluation. The four controls are:

- **No target history:** retain the current image, current target pose, and eight robot states; remove past target poses and their velocity and acceleration features.
- **No future supervision:** disable SE(3), scene-flow and visibility, future depth, DINO, and dynamic-RGB losses; retain current-target RGB reconstruction and the action loss.
- **No scene-flow supervision:** disable only scene-flow and visibility losses; retain dedicated SE(3) and all appearance targets.
- **No dedicated SE(3):** disable only the target-pose forecast loss; retain all appearance targets and the same 14^2 scene-flow queries over the target, robot, and surroundings.

The last control assigns the target the same coarse flow representation as the rest of the scene; it introduces no new target-track head. Since scene flow includes the target and robot, removing it does not isolate environment information. Results are pending: [complete the four controls and their paired intervals before drawing final-scale conclusions].

Screening scope. The broader studies used 4 layers of width 768 and 4 epochs, generally 40,104 updates at global batch size 16. HISTORY1 used a frozen DINO-L encoder and one training seed, with a first-epoch cosine schedule followed by a low learning rate. FUTURE0 used the same depth and width and one seed, but a cosine schedule over all four epochs. Q3-INFO0 used MAE and three seeds for each of R0–R5; BACKBONE0 compared its two backbones with three seeds. The added AUX0 appearance conditions used one seed. These groups are not one matched final-model comparison. Development results motivate design choices; they retain their original split and recipe when cited. The retained screening tables describe completed conditions at their original scale; they do not add 24-layer training arms to the core confirmation.

F.1. Shared Components

The encoder and backbone comparisons belong to the screening scale. They compare a trained CNN, frozen DINOv3 ViT-S/16 and ViT-L/16 [42], and MAE ViT-B/16 [19], with their original group-specific recipes and seed counts. The MAE reference and the BACKBONE0 comparison have three seeds; this does not make every encoder or auxiliary-target condition a three-seed study. The final model uses MAE; DINO-L remains a future feature target. Table S16 retains the screening comparison layout.

Table S16. **Observation-encoder screening layout.** The 4-layer studies retain their source recipes and seed counts (Appendix F.1); the MAE reference has three seeds. Blank cells are unfilled measurements, not a promise of final-scale retraining. Params: encoder parameters; p95: time-to-action.

Encoder	Params	Trained	SR	SR_{150}	p95
CNN	—	yes	—	—	—
DINOv3 ViT-S/16	22M	frozen	—	—	—
DINOv3 ViT-L/16	300M	frozen	—	—	—
MAE ViT-B/16*	86M	frozen	—	—	—

Table S17. **History screening.** Four layers, width 768, frozen DINO-L, one training seed, and the original synchronous evaluation on 1,400 scenes. The first eight conditions receive the current target pose; the final condition adds target-motion history. These are not final-scale real-time results.

History input	Success / 1,400	SR (%)
Current image	748	53.43
RGB window	662	47.29
Optical flow	720	51.43
Flow + motion statistics	723	51.64
Codec motion vectors	733	52.36
Video-VAE latent	710	50.71
Compressed window (age)	634	45.29
Compressed window (uniform)	670	47.86
Current image + target-motion history	824	58.86

F.2. History

HISTORY1 screened current-state input, target motion history, RGB windows, Farnebäck flow [15], flow with motion statistics [33], codec motion vectors, video-VAE latents, and compressed windows. Its nine trained conditions use one seed each. Table S17 reports these nine completed conditions. Only the no-target-history control is repeated at the final scale in the core confirmation.

Stationary-history intervention. The completed intervention in Table S18 uses the frozen 4-layer HISTORY1 checkpoint, one training seed, and the original synchronous executor. It replaces past target poses by the current pose before recomputing kinematics; current pose, image, robot state, timestamps, and validity are retained. The original history result is reused. This intervention changes the input distribution and does not replace the separately trained final-scale no-history control.

Table S18. **Stationary-history intervention.** One frozen 4-layer HISTORY1 checkpoint, training seed 0, and its original synchronous protocol. Past target poses are replaced by the current pose and kinematics recomputed; the unchanged-history result is reused. These 1,400 original scenes are separate from the final real-time confirmation.

Target history	Success / 1,400	SR (%)
Unchanged	824	58.86
Stationary replacement	9	0.64

Pose noise. The separate robustness evaluation freezes the model and reuses the real-time clean reference. It adds three conditions: nominal synthetic temporally correlated pose noise, twice its amplitude, and nominal noise with contiguous missing observations. The perturbation applies to current and historical estimates before history caching and kinematics. Actual tracker calibration and closed-loop tracking remain pending (Table S1). No history-only noise crossover or additional visual-history training is required for this three-condition evaluation. The nominal process combines a fixed per-episode bias and a stationary correlated component, each with per-axis standard deviations of 5 mm and 1° for translation and rotation. The correlation time is 200 ms at 50 ms sampling; the doubled condition scales both components. Dropout lasts four ticks (200 ms), with geometrically distributed valid gaps of mean 76 ticks, giving a long-run missing fraction of 5%. These parameters are set before evaluation. All three final-model training seeds use the same matched scenes at zero added latency; three noise conditions are not three training seeds.

F.3. Forecast

FUTURE0 screened fifteen target configurations with one seed each. Table S19 defines their representations, and Table S20 reports the completed conditions. P09 uses target 2D tracks; it is distinct from the final-scale uniform 3D-flow control. The table also reports AUX0's four MAE appearance-target conditions and their existing all-target reference, all at seed 1. These use a different encoder and recipe from FUTURE0; their scores are compared within each group. They cannot be pooled with Q3's three-seed coupling results or presented as final-scale evidence that geometry outperforms appearance.

Role coverage. The core no-SE(3) control retains the original 14^2 scene flow, and the no-flow control retains SE(3) and all appearance targets. Since the flow covers target, robot, and context, the latter tests scene-flow supervision as a whole. The study makes no claim that a coarse grid matches denser grids.

Extrapolation and forecast errors. Constant-velocity and constant-acceleration estimates from causal pose history can be compared offline with learned forecasts at the logged execution times. Table S21 compares their position errors on the same held-out observations at a fixed 400 ms horizon. CV and CA use only causal pose history; the learned head reads the frozen full-model states. These are offline prediction diagnostics and introduce no controller or training arm. They do not show that the deployed policy reads decoded forecasts. The separately trained CV-input policy remains in Table 1; no additional CV/CA-input training combinations are added.

F.4. Use of the Forecast and Backbone

BACKBONE0 compared shared attention with stream-specific feed-forward networks against separate world/action parameter groups at about 47.3M backbone parameters. Its three-seed results belong to this 4-layer setting.

Q3-INFO0 trained R0–R5 at the same screening scale with three seeds each (Appendix A.5). Its R1 contains world queries with no edge to the actions, whereas the final model reads out from summary tokens without these queries. The Q3 comparisons therefore do not directly measure the final model. Coupling effects changed sign across seeds; this supports the choice of a simpler deployment but does not establish statistical equivalence. R1 minus R0 was [3.14] percentage points on average, with seed-wise differences of [−1.29], [+9.64], and [+1.07]. The core confirmation does not repeat R5 or the other coupling designs at 24 layers.

Deployment check. On the final checkpoint, running unused forecast heads can check equality of the action output for fixed inputs and flow-matching noise and quantify the added computation. Since the real-time executor advances physics during this computation, equal outputs at fixed inputs do not imply equal closed-loop success. No forecast-swap, time-shift, or oracle-input matrix is added for R2/R5.

Table S19. Future-target definitions for the completed screening conditions. Shape excludes batch and query axes. Displacements are cumulative from the current state, except for codec vectors, which keep the correspondence of the decoder. All targets come from the recorded demonstrations. Future depth, DINO, and dynamic RGB are also the training-only auxiliary targets of the selected model.

Target	Shape	Support, units, and validity
Target pose	8×6	Cumulative translation (m) and left-relative rotation vector (rad). Requires a valid pose and query.
Scene flow, 14×14 grid	$196 \times 8 \times 3$	Fixed source grid, cumulative displacement in base-frame meters. Requires valid source depth and a finite future displacement.
Target tracks (2D)	$32 \times 8 \times 3$	Target-mask source points with two image-displacement channels and visibility; padded points are invalid. This is the P09 target, not a 3D rigid-pose fit.
Object DINO	8×1024	Mask-weighted pooling of future patch features with channel layer normalization. Requires a positive mask area.
Scene tracks (2D)	$196 \times 8 \times 3$	Grid source points with two image displacements divided by 224 and a visibility logit.
Dense 2D displacement	$8 \times 2 \times 224 \times 224$	Fixed source pixels, displacement divided by 224. Requires a valid source and a finite projection.
Codec vectors	$8 \times 2 \times 64 \times 64$	Each future/current pair is encoded as P/I frames, and vectors are divided by 64. Only valid vector regions count.
Video-VAE latent	$48 \times 2 \times 14 \times 14$	Frozen Wan2.2 VAE posterior mean with the VAE's own normalization. The two groups end at 200 and 400 ms.
Future depth	$8 \times 1 \times 224 \times 224$	Metric depth on finite positive pixels.
Future DINO	$8 \times 196 \times 1024$	Frozen DINOv3 ViT-L/16 teacher [42], 14×14 patches, channel layer normalization.
Dynamic RGB	$8 \times 3 \times 224 \times 224$	RGB divided by 255 on surfaces moving more than 1 mm, including the robot.

Table S20. **Completed forecast screening.** Both groups use 4 layers of width 768 and 4 epochs, with their original synchronous evaluation. FUTURE0 uses frozen DINO-L and seed 0; its 140 development scenes and disjoint 1,260-scene complement remain separate. AUX0 uses frozen MAE-B, seed 1, and 1,400 original scenes; it retains scene flow, SE(3), and current-target RGB reconstruction. The all-auxiliary MAE reference is reused. Compare conditions within each recipe, not across groups.

ID	Future supervision	Dev. success / 140	Success / 1,260	SR_{1260} (%)
P00	None	88	791	62.78
P01	3D scene flow	95	782	62.06
P02	SE(3)	92	789	62.62
P03	Scene flow + SE(3)	93	786	62.38
P04	Video-VAE latent + SE(3)	92	783	62.14
P05	Dense 2D displacement + SE(3)	87	746	59.21
P06	Codec motion + SE(3)	86	795	63.10
P07	Sparse scene 2D tracks + SE(3)	90	807	64.05
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target 2D tracks	88	808	64.13
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65
AUX0 MAE-B: future appearance auxiliaries		Success / 1,400		SR_{1400} (%)
None		894		63.86
Depth only		867		61.93
DINO only		946		67.57
Dynamic RGB only		926		66.14
All three (existing reference)		995		71.07

Table S21. **Offline prediction errors.** Target-position error at 400 ms (mm) on the same held-out observations. CV and CA are analytic extrapolations from causal pose history; the learned head reads frozen full-model states. These diagnostics are distinct from the trained CV-input policy in Table 1.

	Const. vel.	Harmonic	Roll/slide	Impact
CV extrapolation	—	—	—	—
CA extrapolation	—	—	—	—
Learned pose head	—	—	—	—

Table S22. **Coupling at the screening scale.** Q3-INFO0 used three seeds per condition at 4 layers and width 768. R1 has world queries and is distinct from the final model. SR and p95 retain the source evaluation protocol and hardware; blank cells are unfilled.

Coupling	SR mean	SR std.	p95 (ms)
R0 (future losses off)	—	—	—
R1 (queries, no edge)	—	—	—
R2 (world→action)	—	—	—
R3 (action→world)	—	—	—
R4 (bidirectional)	—	—	—
R5 (decoded geometry)	—	—	—

G. Error Analysis of Acting Under Delay

This appendix derives the error decomposition of Sec. 3.1, extends it to rotation, the context, and chunked execution, derives simple scaling relations, gives magnitudes for Dynamic-14, and states where it does not apply. The decomposition motivates the design of SE(3)-WAM and is not a model of success.

G.1. Translation

Let the true target position at the execution time be $\mathbf{p}(\tau^{\mathrm{a}}) = \mathbf{p}(\tau_t) + \mathbf{v}\Delta + \boldsymbol{\rho}$ and the policy's estimate $\hat{\mathbf{p}}(\tau^{\mathrm{a}}) = \hat{\mathbf{p}}(\tau_t) + \hat{\mathbf{v}}\hat{\Delta} + \hat{\boldsymbol{\rho}}$. Subtracting and adding $\mathbf{v}\hat{\Delta}$ gives the identity

$$\begin{aligned}\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}}) = \big[\hat{\mathbf{p}}(\tau_t) - \mathbf{p}(\tau_t)\big] + (\hat{\mathbf{v}} - \mathbf{v})\hat{\Delta} \\ + \mathbf{v}(\hat{\Delta} - \Delta) + (\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}),\end{aligned} \tag{S4}$$

and the triangle inequality yields Eq. 3. Since $\mathbf{v}$ is the velocity at τ_t, $\boldsymbol{\rho}$ is exactly the part of the motion that constant-velocity extrapolation misses. Under a constant acceleration $\mathbf{a}$, $\boldsymbol{\rho} = \frac{1}{2}\mathbf{a}\Delta^2$, and an acceleration estimate from the history removes part of e_{dyn}. Impacts, rolling contact, and friction changes leave a remainder that only a model of the dynamics can reduce. For a learned predictor, the split of its estimate into $\hat{\mathbf{v}}\hat{\Delta}$ and $\hat{\boldsymbol{\rho}}$ is not identified, because any predicted position can be written in this form. We therefore use the decomposition to say which information each term requires. Our experiments do not separate the terms inside a learned predictor. Instead, they vary the lead, the speed, the pose noise, and the timing, which act on known terms, and they compare variants that differ in the information they receive (Appendix G.5).

G.2. Rotation and the Contact Point

We write $R(\tau^{\mathrm{a}}) = \mathrm{Exp}(\boldsymbol{\omega}\Delta + \boldsymbol{\eta})R(\tau_t)$ and $\hat{R}(\tau^{\mathrm{a}}) = \mathrm{Exp}(\hat{\boldsymbol{\omega}}\hat{\Delta} + \hat{\boldsymbol{\eta}})\hat{R}(\tau_t)$. When R and $\hat{R}$ rotate about a common fixed axis, the exponentials compose additively, and

$$\theta = \big\| \mathrm{Log}(\hat{R}(\tau^{\mathrm{a}})R(\tau^{\mathrm{a}})^{\top})^{\vee} \big\| \le \theta_{\mathrm{loc}} + \hat{\Delta}\, e_{\omega} + \|\boldsymbol{\omega}\|\, e_{\tau} + e_{\eta}, \tag{S5}$$

where θ_{loc} is the current orientation error and e_{ω} and e_{η} are defined as for translation. For general rotations, the Baker–Campbell–Hausdorff formula adds terms of second order in the angles. A contact point at $\mathbf{c}$ in the object frame, with $\|\mathbf{c}\| = r$, has the world position $\mathbf{p} + R\mathbf{c}$, so its error is $(\hat{\mathbf{p}} - \mathbf{p}) + (\hat{R}R^{\top} - I)R\mathbf{c}$. The operator norm of $\hat{R}R^{\top} - I$ for a rotation by θ is $2\sin(\theta/2)$, which gives

$$e_{\mathrm{c}} \le \|\hat{\mathbf{p}} - \mathbf{p}\| + 2r\sin(\theta/2) \le \|\hat{\mathbf{p}} - \mathbf{p}\| + r\theta, \tag{S6}$$

which is the contact bound of Sec. 3.1. Orientation errors therefore enter the contact geometry scaled by the lever arm, and a grasp or placement that depends on orientation needs an SE(3) target state. The planned real-ball protocol uses ball-center translation and excludes target rotation and angular kinematics from inputs, rotation supervision, and rotation errors (Appendix A.2). This task-specific choice leaves robot rotation commands and simulation SE(3) states unchanged; it is not a general symmetry-handling result.

G.3. Context and Clearance

Let $\mathcal{P}$ be the set of context points and times that the executed path passes, and suppose that the path keeps every point $\mathbf{x}$ of the robot and the carried object at least m away from the *predicted* context positions $\hat{\mathbf{y}}_t(\mathbf{z}, \tau)$ for $(\mathbf{z}, \tau) \in \mathcal{P}$. A collision requires $\mathbf{x} = \mathbf{y}_t(\mathbf{z}, \tau)$ for some $(\mathbf{z}, \tau) \in \mathcal{P}$, and then $\|\hat{\mathbf{y}}_t(\mathbf{z}, \tau) - \mathbf{y}_t(\mathbf{z}, \tau)\| = \|\hat{\mathbf{y}}_t(\mathbf{z}, \tau) - \mathbf{x}\| \ge m$, so a collision requires a prediction error of at least m somewhere along the path. Context precision finer than m thus has no first-order effect on success. The bound concerns the size of the error and says nothing about sampling. A flow sampled on a grid misses structures thinner than its spacing and errs at motion boundaries. Both effects enter the path error e_F, which cannot be inferred from grid spacing alone. The margin is itself a design variable, since a policy can keep a larger margin when its context estimate is coarse, at some cost in reachability or time.

G.4. Leads Under Rolling Replanning

Under the real-time executor of Appendix C.3, query t has total latency ℓ_t, including any added delay. Its chunk is adopted after $k_t\Delta_a$, with $k_t = \max(1, \lceil \ell_t/\Delta_a \rceil)$, and the next query starts at that same boundary. For chunk length $K = 8$, commands that reach execution have leads

$\Delta_{t,j} = (k_t + j)\Delta_a + d_{\text{task}}$, $j = 0, \ldots, \min(K, k_{t+1}) - 1$, before replacement or exhaustion. Here d_{task} is the built-in command delay: 50 ms for post-impact grasp and zero otherwise. Holding after exhaustion adds no new command. With $k_t = k_{t+1} = k \leq 4$ (total latency at most 200 ms), the largest lead is $(2k-1)\Delta_a + d_{\text{task}}$ and remains inside the 400 ms forecast, including the task delay. At larger total delays, the last commands can execute beyond this horizon. The full model's action stream does not receive the latency, so it does not adapt its commands to a particular delay. Replanning at the first eligible boundary keeps the leads short; the latency sweep measures the effect of longer leads on success.

G.5. Scaling Relations

The decomposition gives the following reference relations. Fig. S6 reports motion and synthetic-noise diagnostics.

Critical lead. A policy that does not compensate motion ($\hat{\mathbf{v}} = 0$, $\hat{\boldsymbol{\rho}} = 0$) incurs $\|\mathbf{v}\|\Delta$ in addition to e_{loc} and $\|\boldsymbol{\rho}\|$. With a small localization error and smooth motion, its contacts fail once $\|\mathbf{v}\|\Delta > \varepsilon$, i.e., beyond the critical lead $\Delta^\star = \varepsilon/\|\mathbf{v}\|$, and its success depends on the task, the speed, and the latency only through $\|\mathbf{v}\|\Delta/\varepsilon$. Table S5 lists ε, $\|\mathbf{v}\|$, and $\Delta^\star$ for every task. Fig. S6a plots the success of each method against $\|\mathbf{v}\|\Delta/\varepsilon$, computed per episode at the first command's execution time and pooled over the existing task and latency conditions, together with the Spearman correlation and a fitted logistic curve for the reactive policies. The reactive policies [fall on one curve] (Spearman $\rho = [X.XX]$), and their success halves at [X.X]. The decomposition gives only a threshold of about one, not the shape of this curve. The capture tasks scatter most around it, because their success also depends on when the gripper closes.

Pose noise. A least-squares velocity estimate from n equally spaced poses that span T, each with independent noise σ per axis, has the standard deviation $\sigma\sqrt{12(n-1)/(n(n+1))}/T \approx \sigma\sqrt{12/n}/T$ [5]. A pose history therefore keeps $\hat{\Delta}e_v$ below the tolerance up to $\sigma^\star = \varepsilon T\sqrt{n}/(\Delta\sqrt{12})$. For $\varepsilon = 10$ mm, $T = 350$ ms, $n = 8$, and $\Delta = 200$ ms, this gives $\sigma^\star \approx 14$ mm. A learned estimator with a motion prior can do better, so $\sigma^\star$ is the value for a model-free linear estimate under independent noise. Our robustness test instead uses three synthetic conditions: temporally correlated noise, twice its amplitude, and nominal noise with contiguous missing observations. It reuses the clean real-time reference and does not estimate a noise crossover or claim calibration to a physical tracker.

Clearance. A context forecast with path error e_F avoids collisions while $m > e_F$, under the assumptions of Appendix G.3. This is a geometric sufficient condition, not an empirical equivalence claim between coarse and fine flow grids.

Figure placeholder: Fig. S6 — uncompensated motion and pose noise

(a) x: $\|\mathbf{v}\|\Delta/\varepsilon$ at the first command's execution time (log); y: SR. One point per method × family × latency, reusing the main latency sweep; reactive policies (gray) with a fitted logistic curve and Spearman ρ, the threshold at 1 marked; motion-compensating baselines and SE(3)-WAM (orange). **(b)** x: clean reference, nominal correlated noise, twice the amplitude, and nominal noise with contiguous missing observations; y: SR. Frozen weights and the same real-time protocol; paired intervals from matched scenes.

Figure S6. **Motion and pose-noise diagnostics.** (a) Success against uncompensated motion relative to the tolerance, comparing reactive and motion-compensating methods. (b) Robustness under the three synthetic pose-noise conditions, relative to the clean reference.

G.6. Magnitudes in Dynamic-14

Table S23 evaluates the terms for representative Dynamic-14 settings (Table S6). Each of them is comparable to or larger than the millimeter-scale tolerances of the success conditions (Table S4).

Table S23. Illustrative magnitudes of the terms of Eq. 3 for a 200 ms lead. The values follow from the stated quantities and are not measurements.

Term	Setting	Size
$\|\mathbf{v}\|\Delta$	conveyor at 0.10 m/s	20 mm
$\hat{\Delta}e_v$	10% speed misestimate	2.8 mm
$\|\mathbf{v}\|e_\tau$	50 ms timing error (one period)	7.0 mm
$2r\sin(\theta/2)$	$\omega = 1.2$ rad/s unpredicted, $r = 3$ cm	7.2 mm
e_{dyn}	ball on a 10° ramp, no acceleration model	24 mm
ε	grasp of the conveyor part	[X] mm
grid spacing	14 × 14 flow at the workspace	[X] mm

The ramp entry uses the acceleration of a solid sphere that rolls without slipping, $a = \frac{5}{7}g\sin\alpha \approx 1.2\,\text{m/s}^2$ at $\alpha = 10°$, and $\frac{1}{2}a\Delta^2$ for $\Delta = 0.2$ s.

G.7. Scope

The decomposition addresses the approach to a rigid target whose motion is exogenous until contact. It does not cover deformable or articulated targets, whose motion field on $\mathcal{O}$ is not determined by six numbers. It also does not cover the motion after contact, which depends on the action itself and calls for action-conditioned prediction, the case in which letting the world read the action could matter. Scenes in which the target's role changes over time (e.g., sequential multi-object tasks) are outside its scope as well. Perception

quality enters only through e_{loc} and e_v, and we measure it separately (Table S1). The bound is not a probability of success. It identifies which errors scale with which factors.

H. Additional Related Work

This appendix extends Sec. 2 with work that the main text covers only briefly.

Dynamic manipulation and latency. Classical systems compensate latency with predictive tracking [4] and visual servoing [23], and catch or intercept moving objects by planning on estimated trajectories [2, 6, 20], later combined with grasp and reachability planning [1, 34]. Learned visuomotor policies [12, 57] and vision–language–action models [8, 9, 27, 37, 38, 62] generalize across objects and instructions, and most of them are trained and evaluated on scenes that stay still while the model computes. Among learned methods for moving scenes, DynamicManip synthesizes dynamic demonstrations from a single static one [28], and DynaWM refines the action chunk of a base VLA for moving objects [10].

World models and world–action models. World models have been used for planning with visual foresight [16], for reinforcement learning in a learned latent space [17, 18], and for planning on pretrained visual features [3, 59]. In manipulation, generated frames or subgoal images have been converted into actions [7, 13, 56]. Studies of WAMs compare how the video and action streams should exchange information [47, 52] and how long and how compressed their context should be [22, 32]. Further forecast targets include motion-weighted video [24] and optical flow or motion vectors [11, 29].

3D and motion representations. Object-centric point clouds improve generalization [61], and 3D and 4D VLAs bring 3D and spatiotemporal inputs to large models [54, 58]. Dense pixel motion has also been predicted as an intermediate representation [36, 39, 40]. Beyond SE(3) decompositions, 3D dynamic scene memories [51] and scene flow estimation [31, 43] model how scenes move. Point trackers, pose estimators, and feed-forward reconstruction models [25, 45, 48, 49, 53] extract such motion labels from video; our DROID pretraining uses the 3D point tracks that PointWorld extracted [21] (Appendix B).

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